Solution Architecture Document

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1. Architecture Overview

JobMatch is an event-driven, service-oriented system. A single API gateway fronts a set of small Python services; long-running work (CV parsing, matching, notifications, partner synchronisation) is handled asynchronously through a message broker. Every service is stateless and runs as a container on Kubernetes. Architecture style: service-oriented, event-driven, deployed as containers on Kubernetes.

2. Technology Stack

Layer	Technology
Web client	React 18, TypeScript, Vite
Services	Python 3.12, FastAPI, Celery workers
Machine learning	scikit-learn and sentence-transformers embeddings, served by the Matching Engine
Relational data	PostgreSQL 16
Search	OpenSearch 2.13
Cache and sessions	Redis 7
Messaging	RabbitMQ 3.13
Object storage	MinIO (S3 compatible)
Identity	Keycloak 24 (OpenID Connect)
Platform	Kubernetes 1.30, NGINX Ingress, Helm 3
Infrastructure as code	Terraform 1.8, Ansible 9
CI/CD	GitLab CI with a private container registry
Observability	Prometheus, Grafana, Loki, Alertmanager

3. Components

Component	Purpose	Technology	Owner	Depends on
Web Frontend	Single-page application for job seekers, employers and staff	React 18, TypeScript	Frontend Team	API Gateway, Identity Service

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API Gateway	Single entry point: routing, rate limiting, token validation	NGINX Ingress, Kong	Platform Team	Identity Service
Identity Service	Sign-in, single sign-on, roles and token issuing	Keycloak 24	Platform Team	PostgreSQL
Profile Service	Job seeker profiles, CV metadata and consent records	Python 3.12, FastAPI	Core Services Team	PostgreSQL, MinIO, RabbitMQ
Employer Service	Employer accounts, vacancies and applications	Python 3.12, FastAPI	Core Services Team	PostgreSQL, RabbitMQ
Resume Parser	Extracts skills and experience from uploaded CVs	Python 3.12, Celery, spaCy	Matching Team	MinIO, RabbitMQ
Matching Engine	Computes and ranks matches between profiles and vacancies	Python 3.12, scikit-learn, sentence-transformers	Matching Team	OpenSearch, PostgreSQL, Redis
Notification Service	Sends e-mail and SMS notifications	Python 3.12, Celery	Core Services Team	RabbitMQ, SMS gateway, e-mail relay
Partner Sync Service	Synchronises vacancies with partner portals	Python 3.12, Celery	Integrations Team	Employer Service, jobs.ps API, RabbitMQ
Reporting Service	Aggregates statistics and exports reports	Python 3.12, pandas	Data Team	PostgreSQL
Admin Console	Back-office screens for staff and administrators	React 18, TypeScript	Frontend Team	API Gateway

4. Data Stores

Data store	Type	Technology	Contents
Primary database	Relational database	PostgreSQL 16 (primary with two replicas)	Profiles, vacancies, applications, consent records, audit schema
Search index	Search engine	OpenSearch 2.13 (three nodes)	Searchable vacancy and candidate documents with embeddings
Session cache	In-memory cache	Redis 7 (sentinel)	Sessions, rate-limit counters, match result cache
Message broker	Message queue	RabbitMQ 3.13 (three-node cluster)	Work queues for parsing, matching, notifications and sync
Document store	Object storage	MinIO (erasure coded)	Uploaded CV files and generated report exports

5. Integrations

Integration	Direction	Protocol	Purpose	Data exchanged	Authentication
jobs.ps partner portal	Bidirectional	REST over HTTPS, JSON	Publish vacancies and read application status	Vacancy records and status events	OAuth 2.0 client credentials
National Civil Registry	Outbound	SOAP over HTTPS	Verify the national ID of a registering job seeker	National ID number and date of birth; yes/no result	Mutual TLS with a client certificate
Labour Market Information System (LMIS)	Outbound	REST over HTTPS, JSON	Monthly statistics exchange	Aggregated placement figures, no personal data	OAuth 2.0 client credentials
SMS gateway	Outbound	REST over HTTPS	Send SMS notifications	Phone number and message text	API key in the Authorization header
E-mail relay	Outbound	SMTP over TLS	Send e-mail notifications	Recipient, subject and body	SMTP authentication with a service account
Skills taxonomy service	Inbound	REST over HTTPS, JSON	Fetch the standard skills list used by the parser	Skill names and codes	API key

6. Security Architecture

- All traffic is encrypted with TLS 1.2 or higher; internal service traffic uses mutual TLS inside the cluster.

- Personal data is encrypted at rest (database volumes, object storage).

- Access is role based and every change to a profile is written to the audit schema.

7. Diagram Reference

The component diagram is stored in the documentation repository at docs/architecture/jobmatch-components-v2.1.drawio.